第 4 阶段。记录信息并停用 node2

AFF and FAS Controller Upgrade

NetApp
January 17, 2022

This PDF was generated from https://docs.netapp.com/zh-cn/ontap-systems-upgrade/upgrade-arl-manual/stage_4_index.html on January 17, 2022. Always check docs.netapp.com for the latest.

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第 4 阶段。记录信息并停用 node2

第 4 阶段。记录 node2 信息并停用 node2

在阶段 4 中，您会记录 node2 信息，然后停用 node2 。

步骤

1. "记录 node2 信息"
2. "停用 node2"

记录 node2 信息

在关闭并停用 node2 之前，您需要记录有关其集群网络，管理和 FC 端口以及 NVRAM 系统 ID 的信息。稍后将 node2 映射到 node4 并重新分配磁盘时，您需要在操作步骤中提供此信息。

步骤

1. 在 node2 上查找集群网络，节点管理，集群间和集群管理端口：

   ```
   network interface show -curr-node <node_name> -role cluster ,  intercluster ,
   nodemgmt ,  cluster-mgmt
   ```

   系统将显示该节点以及集群中其他节点的 LIF ，如以下示例所示：

   ```
   cluster::> network interface show -curr-node node2 -role
   cluster,intercluster,node-mgmt,cluster-mgmt
            Logical     Status     Network            Current     Current
   Is
   Vserver  Interface   Admin/Oper Address/Mask       Node        Port
   Home
   -------- ----------  ---------- ------------------ ----------  -------
   ----
   node2
            intercluster up/up     192.168.1.202/24   node2       e0e
   true
            clus1        up/up     169.254.xx.xx/24   node2       e0a
   true
            clus2        up/up     169.254.xx.xx/24   node2       e0b
   true
            mgmt1        up/up     192.168.0.xxx/24   node2       e0c
   true
   4 entries were displayed.
   ```

> 您的系统可能没有集群间 LIF 。您只能在一个节点对的一个节点上使用集群管理 LIF 。的示例输出显示了集群管理 LIF "第 1 步" 在 *Record node1 port information*.

2. 捕获输出中的信息以在部分中使用 "将端口从 node2 映射到 node4"。

   要将新控制器端口映射到旧控制器端口，需要提供输出信息。

3. 确定节点 2 上的物理端口：

   `network port show -node <node_name> -type physical +`

   `node_name` 是要迁移的节点。

   系统将显示 node2 上的物理端口，如以下示例所示：

```
cluster::> network port show -node node2 -type physical

                                                                  Speed
(Mbps)
Node   Port      IPspace      Broadcast Domain Link   MTU    Admin/Oper
------ --------- ------------ ---------------- ----- ------- -----------
node2
       e0M       Default      IP_address       up       1500  auto/100
       e0a       Default      -                up       1500  auto/1000
       e0b       Default      -                up       1500  auto/1000
       e1a       Cluster      Cluster          up       9000  auto/10000
       e1b       Cluster      Cluster          up       9000  auto/10000
5 entries were displayed.
```

4. 记录端口及其广播域。

   稍后，需要在操作步骤中将广播域映射到新控制器上的端口。

5. 确定 node2 上的 FC 端口：

   `network fcp adapter show`

   系统将显示 node2 上的 FC 端口，如以下示例所示：

```
cluster::> network fcp adapter show -node node2
                      Connection  Host
Node          Adapter Established Port Address
------------ ------- ----------- ------------
node2
              0a      ptp         11400
node2
              0c      ptp         11700
node2
              6a      loop        0
node2
              6b      loop        0
4 entries were displayed.
```

6. 记录端口。

   稍后在操作步骤中映射新控制器上的新 FC 端口需要输出信息。

7. 如果您之前没有这样做，请检查 node2 上是否配置了接口组或 VLAN：

   ```
   ifgrp show
   ```

   ```
   VLAN show
   ```

   您将使用部分中的信息 "将端口从 node2 映射到 node4"。

8. 执行以下操作之一：

如果您 ...	那么 ...
已在中记录 NVRAM 系统 ID 编号 "准备要升级的节点"	转至 "停用 node2"。
未在中记录 NVRAM 系统 ID 编号 "准备要升级的节点"	完成 第 9 步 和 第 10 步 然后转到下一节， "停用 node2"。

9. 【man_record_2_step9]] 显示节点 2 的属性：

   ```
   ssystem node show -instance -node node2
   ```

   ```
   cluster::> system node show -instance -node node2
                                ...
                   NVRAM System ID: system_ID
                               ...
   ```

10. 【man_record_2_step10]] 记录要在部分中使用的 NVRAM 系统 ID "安装并启动节点 4."。

停用 node2

要停用 node2 ，您需要正确关闭 node2 并将其从机架或机箱中卸下。如果集群位于 SAN 环境中，则还需要删除 SAN LIF 。

步骤

1. 执行以下操作之一：

如果集群 ...	那么 ...
双节点集群	转至 第 2 步。
包含两个以上节点的集群	转至 第 9 步。

2. 【 man_REALET_2_Step2]] 在任一节点上输入以下命令，以访问高级权限级别：

   ```
   set -privilege advanced
   ```

3. 输入以下命令并检查其输出，以验证是否已禁用集群 HA ：

   ```
   cluster ha show
   ```

   系统将显示以下消息：

   ```
   High Availability Configured: false
   ```

4. 输入以下命令并检查其输出，以检查 node2 当前是否持有 epsilon ：

   ```
   cluster show
   ```

   以下示例显示 node2 持有 epsilon ：

   ```
   cluster*::> cluster show
   Node                  Health  Eligibility   Epsilon
   -------------------- ------- ------------  -----------
   node1                 true    true          false
   node2                 true    true          true

   Warning: Cluster HA has not been configured. Cluster HA must be
   configured on a two-node cluster to ensure data access availability in
   the event of storage failover. Use the "cluster ha modify -configured
   true" command to configure cluster HA.

   2 entries were displayed.
   ```

5. 如果 node2 持有 epsilon ，请在节点上将 epsilon 标记为 `false` ，以便可以将其传输到 node3 ：

```
cluster modify -node <node2> -epsilon false
```

6. 将 epsilon `true` 标记为 node3 ，将 epsilon 传输到 node3 ：

```
cluster modify -node <node3> -epsilon true
```

7. 验证设置是否为双节点无交换机集群：

```
network options switchless-cluster show
```

```
cluster::*> network options switchless-cluster show
Enable Switchless Cluster: false/true
```

此命令的值必须与系统的物理状态匹配。

8. 验证设置是否为双节点无交换机集群：

```
network options switchless-cluster show
```

```
cluster::*> network options switchless-cluster show
Enable Switchless Cluster: false/true
```

此命令的值必须与系统的物理状态匹配。

9. 返回到管理员级别：

```
set -privilege admin
```

10. 在任一控制器上输入以下命令以暂停 node2： `ssystem node halt -node node2`
11. node2 完全关闭后，将其从机箱或机架中卸下。升级完成后，您可以停用 node2 。请参见 "停用旧系统"。

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